Examiner's Use

6 **(a)** A stationary nucleus of a radioactive nuclide, ${}^{218}_{84}Po$, underwent a chain of decays by the emission of an α and β–particles. The decay is represented by the two equations:

$$ {}^{218}_{84}Po \rightarrow D + \alpha $$
$$ D \rightarrow E + \beta $$

where D and E are the nuclides formed after the decay.

(i) State the nuclear notation for E. [1]

$$ {}^{\cdots\cdots}_{\cdots\cdots}E $$

(ii) Determine the ratio of the kinetic energy of the α–particle to the total kinetic energy of D and α–particle.

ratio = …… [3]

For Examiner's Use

(iii) In reality, the β –particles have a range of kinetic energies, instead of a fixed value. Explain why this is so.

……..

……..

……..…[1]

(iv) A sample of $^{218}_{84}Po$ is placed on a weighing balance and a reading of 4.05 g is obtained. After 243 s, the reading drops to 4.02 g.

1. Determine the number of particles $^{218}_{84}Po$ in the initial sample.

number of particles = …….. [1]

2. Show that the total number of particles D and E after 243 s is 4.52 x 10^{21}. [1]

3. Determine the half life of $^{218}_{84}Po$.

half life = ……. s [2]

Examiner's Use

(b) For many unstable parent nuclei, the daughter nuclei is itself radioactive. This may give rise to a radioactive series where there may be ten or more different radioactive daughter products.

A radioactive parent nucleus X has a radioactive daughter nucleus Y and, in turn, this daughter produces a further stable daughter Z. The variation with time *t* of the percentage number *P* of the different nuclei X, Y and Z in a radioactive sample is illustrated in Fig. 6.1

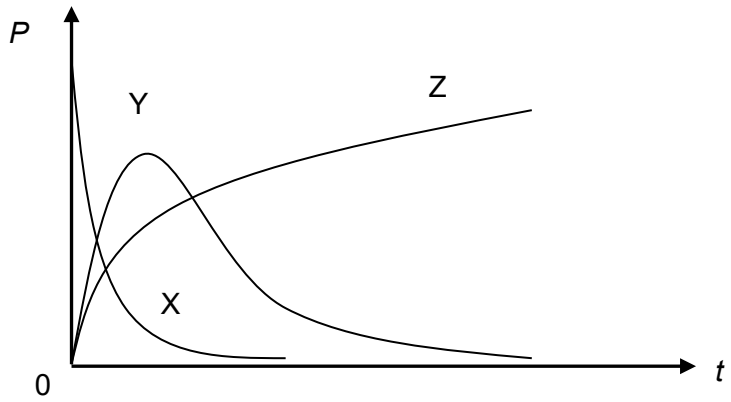

Fig 6.1

(i) X has a half life of 5 hours. The count rate at $t = 0$ and $t = 5$ hours were measured. Suggest three possible reasons why the count rate is not exactly halved after 5 hours.

1. ……

……

2. ……

……

3. ……

……[3]

(ii) Explain why graph of Y increases to a maximum and then decreases.

……

……

……

……[2]